

DATA BASE SYSTEMS ARCHITECTURE

Databases III

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 DataBases Infrastructure 
- 2 DBMS Architecture 
- 3 Database System Administration 
- 4 Transactional System 
- 5 Query Execution 
- 6 Concurrency Control 
- 7 Failure Recovery 



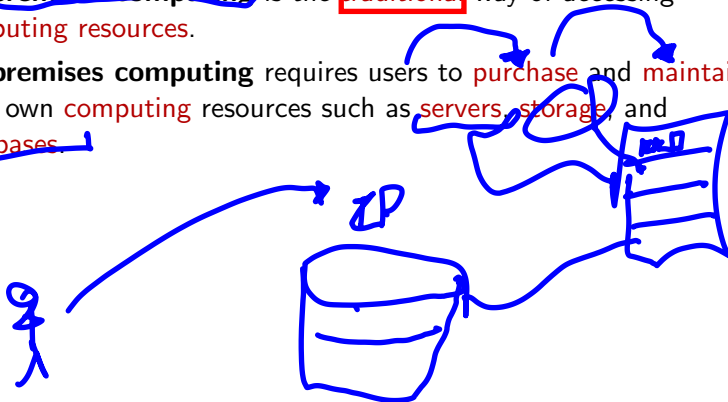
Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



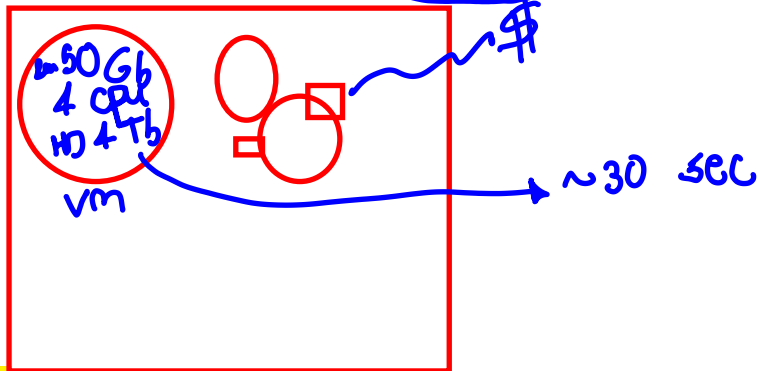
What is On-Premises Computing?

- **On-premises computing** is the **traditional** way of accessing computing resources.
- **On-premises computing** requires users to **purchase** and **maintain** their own **computing** resources such as **servers**, **storage**, and **databases**.



What is the Cloud Computing?

- **Cloud computing** is the delivery of **computing services** over the internet.
- **Cloud computing** allows users to **access computing resources** such as servers, storage, and databases **on demand**.



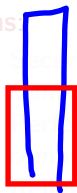
Pros & Cons of Cloud Computing

• Pros:

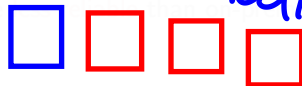
- **Cost-Effective:** Cloud computing is a cost-effective way to access computing resources.
- **Scalable:** Cloud computing is a scalable way to access computing resources.
- **Flexible:** Cloud computing is a flexible way to access computing resources.

• Cons:

- **Security:** Cloud computing can be less secure than on-premises computing.
- **Availability:** Cloud computing can be less available than on-premises computing.
- **Performance:** Cloud computing can be slower than on-premises computing.
- **Cost:** Cloud computing can be more expensive than on-premises computing.



vertical



horizontal



Pros & Cons of Cloud Computing

● Pros:

- **Cost-Effective:** Cloud computing is a **cost-effective** way to access computing resources.
- **Scalable:** Cloud computing is a **scalable** way to access computing resources.
- **Flexible:** Cloud computing is a **flexible** way to access computing resources.

● Cons:

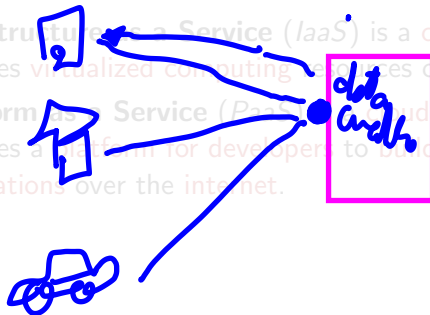
- **Security:** Cloud computing can be **less secure** than on-premises computing.
- **Performance:** Cloud computing can be **slower** than on-premises computing.
- **Reliability:** Cloud computing can be **less reliable** than on-premises computing.

network



SaaS Vs. IaaS Vs. PaaS

- **Software as a Service (SaaS)** is a **software distribution** model in which a **third-party** provider **hosts applications** and makes them available to customers over the **internet**.



SaaS Vs. IaaS Vs. PaaS

- **Software as a Service (SaaS)** is a **software distribution** model in which a **third-party** provider **hosts applications** and makes them available to customers over the **internet**.
- **Infrastructure as a Service (IaaS)** is a **cloud computing** model that provides **virtualized computing** resources over the **internet**.
- **Platform as a Service (PaaS)** is a **cloud computing** model that provides a **platform** for developers to **build, deploy, and manage** applications over the **internet**.

zac → terraform



SaaS Vs. IaaS Vs. PaaS

- **Software as a Service (SaaS)** is a **software distribution** model in which a **third-party** provider **hosts applications** and makes them available to customers over the **internet**.
- **Infrastructure as a Service (IaaS)** is a **cloud computing** model that provides **virtualized computing** resources over the **internet**.
- **Platform as a Service (PaaS)** is a **cloud computing** model that provides a **platform for developers** to **build, deploy, and manage** applications over the **internet**.

/ \$ rent

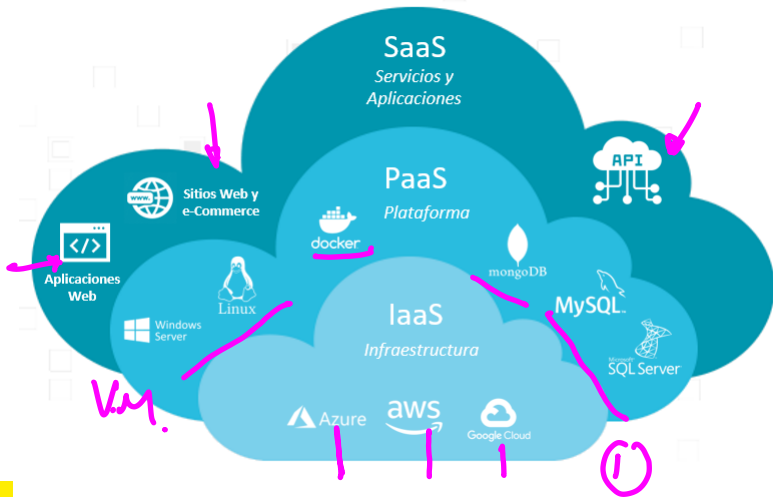
no admin



Docker - K8s
Artifacts
ML/AI



Cloud Levels



DataBases as a Service

Database as a Service (DBaaS) is a cloud computing model that provides database services over the internet.

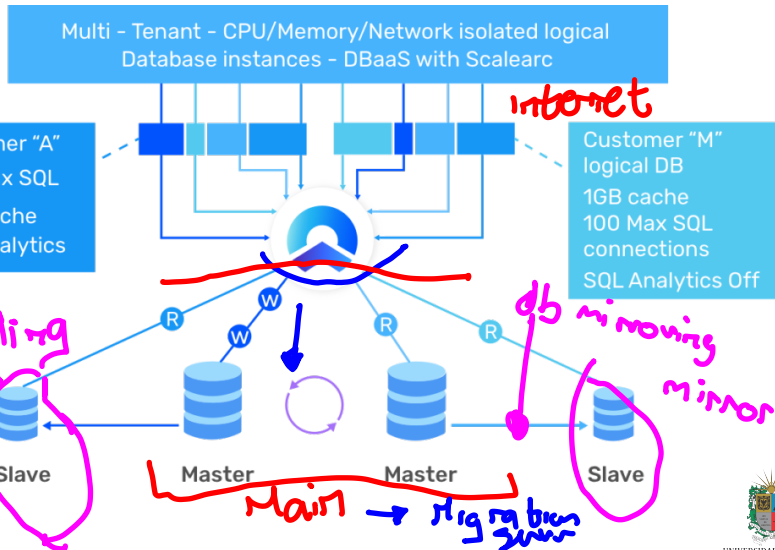
↑
 ① DBMS $\begin{cases} \text{NoSQL} \\ \text{SQL} \end{cases}$ } ready to use

② Security: Firewall, Duty, Logs, Policies

③ Migration (Snowflake $\begin{cases} \text{mobile} \\ \text{ball} \end{cases}$)



Case Study: DBaaS Custom for Clients



Outline

1 Databases Infrastructure

2 DBMS Architecture

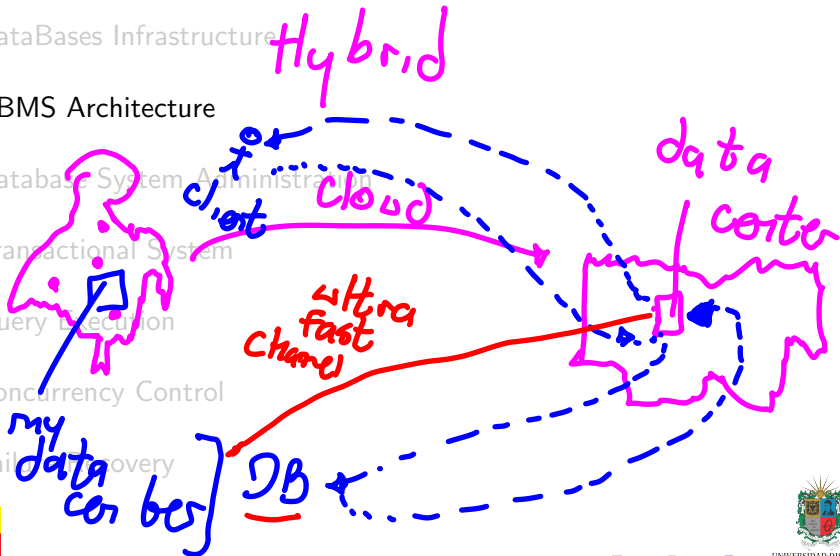
3 Database System Administration

4 Transactional System

5 Query Execution

6 Concurrency Control

7 Failure Recovery



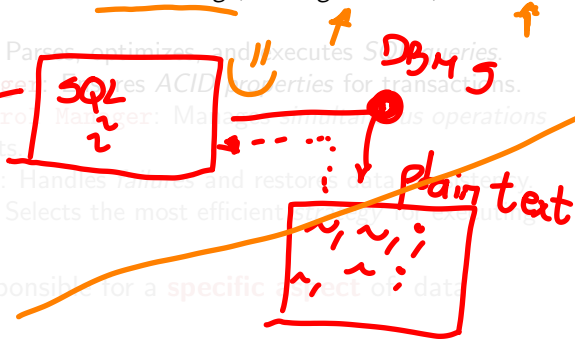
DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles data storage, file organization, and access methods.

- **Query Processor**: Parses, optimizes, and executes SQL queries.
- **Transaction Manager**: Enforces ACID properties for transactions.
- **Concurrency Control Manager**: Manages simultaneous operations and prevents conflicts.

- **Recovery Manager**: Handles failures and restores data to a consistent state.
- **Optimizer**: Selects the most efficient SQL query for a given query.

MySQL
PostgreSQL
Oracle



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes SQL queries.
 - Handwritten annotations:
 - graph** (pink) with an arrow pointing to "Parses"
 - tree** (pink) with an arrow pointing to "Parses"
 - compiler** (orange) with an arrow pointing to "optimizes"
 - language to interact with storage** (blue) with an arrow pointing to "executes"
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage, file organization, and access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient strategy for executing queries.
- Each component is responsible for a **specific aspect** of data management.

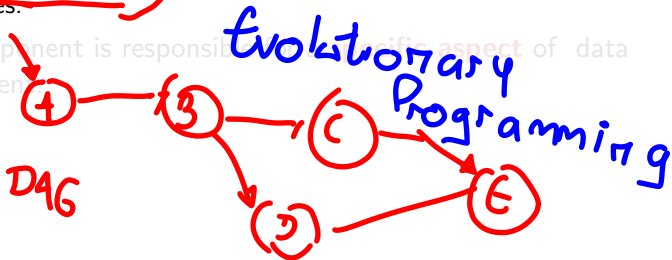
hand
inherent
Backup



DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.

- Each component is responsible for a specific aspect of data management.



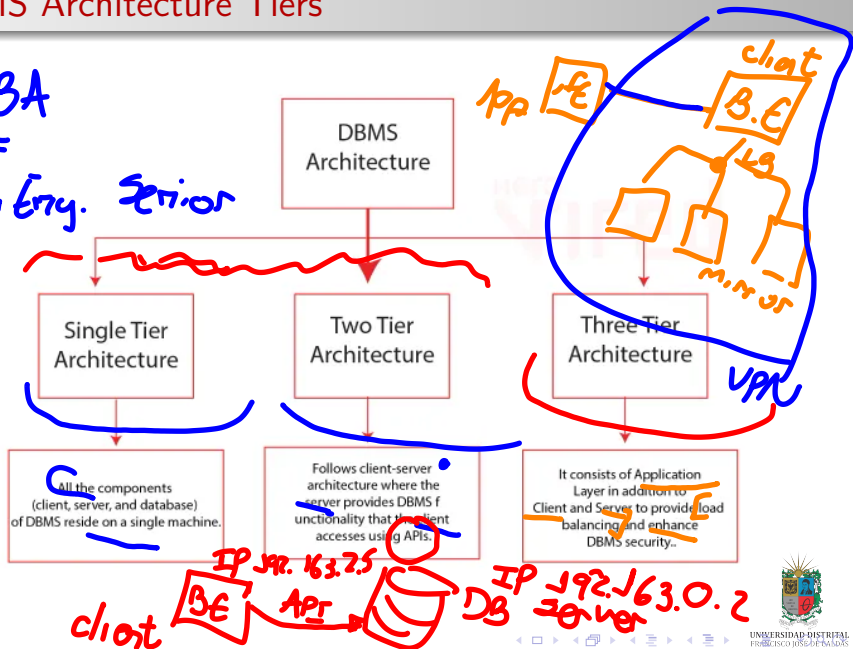
DBMS Architecture Overview

- A **Database Management System (DBMS)** is organized in **layers**:
 - **Storage Manager**: Handles *data storage*, *file organization*, and *access methods*.
 - **Query Processor**: Parses, optimizes, and executes *SQL queries*.
 - **Transaction Manager**: Ensures *ACID properties* for transactions.
 - **Concurrency Control Manager**: Manages *simultaneous operations* and prevents conflicts.
 - **Recovery Manager**: Handles *failures* and restores data consistency.
 - **Query Optimizer**: Selects the most efficient *strategy* for executing queries.
- Each component is responsible for a **specific aspect** of data management

a/1



DBA
=
Data Entry Senior



DBMS Architecture N-Tier

DBMS Architecture



DatabaseTown.com

It is responsible for providing an interface for users.

It is responsible for managing the different views of the data in the database.

It is responsible for managing the logical organization of data in the database.

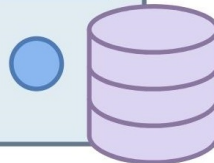
It is responsible for managing the physical storage of data on disk.

Application
Layer

View
Layer

Logical
Layer

Physical
Layer



Backend

Relational
Algebra

Form (write
(INSERT))

GUI report
screen
(SELECT)

Component



Types of DBMS Architecture

There are several types of DBMS architectures:

- **Centralized DBMS:** All components are on a single server.
- **Client-Server DBMS:** Clients access the database through a server.
- **Distributed DBMS:** Data is distributed across multiple servers.
- **Cloud DBMS:** Database services are provided over the cloud.
- **Hybrid DBMS:** Combines features of centralized and distributed systems.
- **Peer-to-Peer DBMS:** Each node can act as a client and server.
- **In memory DBMS:** Data is stored in RAM for faster access.

1-tier
developers



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services provided over the cloud.
- **Hybrid DBMS**: Combines features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a client and server.
- **In memory DBMS**: Data is stored in **RAM** for faster access.

internet

backend
|
middleware

own
IP



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a client and server.
- **In memory DBMS**: Data is stored in **RAM** for faster access.

parallelism

no replicas



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: Combines features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node acts as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.

Handwritten notes: "mix" with arrows pointing to "rel" and "dist", and "ser".



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS** Combines features of centralized and distributed systems.
 - **Peer-to-Peer DBMS**: Each node can act as a client and server. *on-premise*
 - **In memory DBMS**: Data is stored in **RAM** for faster access. *cloud*



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.



Types of DBMS Architecture

There are several types of **DBMS architectures**:

- **Centralized DBMS**: All components are on a **single server**.
- **Client-Server DBMS**: Clients **access** the database through a **server**.
- **Distributed DBMS**: Data is distributed across **multiple servers**.
- **Cloud DBMS**: Database services are provided over the **cloud**.
- **Hybrid DBMS**: **Combines** features of centralized and distributed systems.
- **Peer-to-Peer DBMS**: Each node can act as a **client** and **server**.
- **In memory DBMS**: Data is stored in **RAM** for faster access.

DB - 100-200ms
Cache - 5-10 ms

Cache
lifetime (sec-min) max 30 mins



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration**
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** database systems to ensure their reliability, performance, security, and availability.

- Key responsibilities include:

Data storage

and

everything
it works

memory
processor
↓
time

profiling

grants
backups

accessible
scalability

- Database administrators (DBAs) are essential for modern organizations.



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** database systems to ensure their reliability, performance, security, and availability.
- Key responsibilities include:

- **Data storage management**

- System architecture design

- Transaction and query processing

- Concurrency control

- Security and recovery

- Storage access control

- Database administrators (DBAs) are essential for modern organizations

Files
(PDFs)

- Data Lake
- Knowledge Bases

NoSQL

SQL

Dashboards
analytics

cloud?

no frequent legal

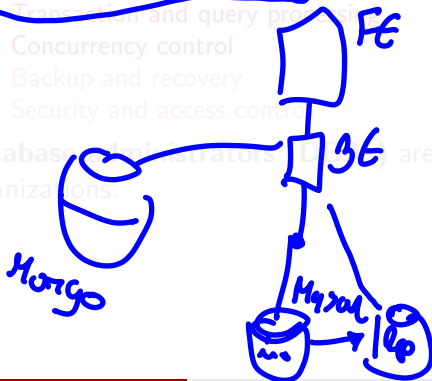
cache?

frequent
no change



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing

→ **Indexes**

→ **Improve DB model**

→ **Internal Optimization**



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
 - Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
 - Database administrators (DBAs) are essential for modern organizations.
- Handwritten notes:* A blue bracket under "Concurrency control" points to "sync". Another blue arrow points from "sync" to "replicas / only read".



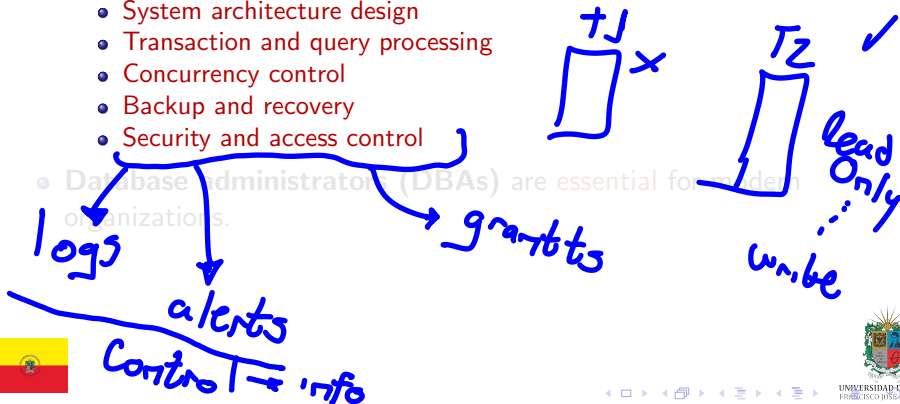
- ## Incremental

Incremental today



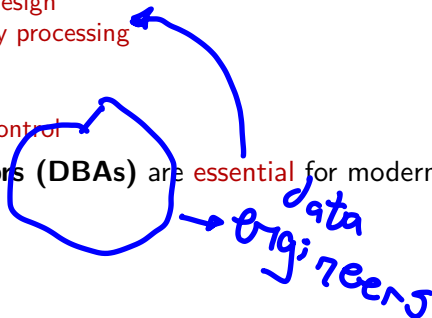
Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control



Database System Administration

- **Database system administration** is the discipline of **managing**, **configuring**, and **maintaining** *database systems* to ensure their reliability, performance, security, and availability.
- Key responsibilities include:
 - Data storage management
 - System architecture design
 - Transaction and query processing
 - Concurrency control
 - Backup and recovery
 - Security and access control
- **Database administrators (DBAs)** are **essential** for modern organizations.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a database table.

- Efficient record storage is crucial for fast data retrieval and update.

- Storage techniques:
 - Heap storage: simple, fast inserts.
 - Ordered storage: fast range queries.
 - B-trees: fast lookups.

save variables

r1	...
r2	...
r3	...

[{ r1
},
{ r2
},
{ r3
}]



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - Heap storage: Unordered, fast inserts.
 - Sequential storage: Ordered, fast range queries.
 - B-tree storage: Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- **Storage techniques:**
 - **Heap storage:** Unordered, fast inserts.
 - **Sequential storage:** Ordered, fast range queries.
 - **Indexed storage:** Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - **Heap storage**: Unordered, fast inserts.
 - **Sequential storage**: Ordered, fast range queries.
 - **Indexed storage**: Uses indexes for fast lookups.



Record Storage Concepts

- A **record** (or row/tuple) is the **basic unit** of data storage in a *database table*.
- **Efficient record storage** is crucial for **fast data retrieval** and **update**.
- Storage techniques:
 - **Heap storage**: Unordered, fast inserts.
 - **Sequential storage**: Ordered, fast range queries.
 - **Indexed storage**: Uses indexes for fast lookups.

sort / binary search



Block and Page Organization

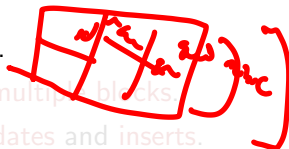
- Data is stored in **blocks** (pages) on disk.
- Block size and layout affect I/O performance.
- Records may be packed, slotted, or may span multiple blocks.
- Free space management is important for updates and inserts.
- Fragmentation management ensures efficient use of space.
- Data compression can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.

- Records may be packed, slotted, or may span multiple blocks.
- Free space management is important for updates and inserts.



Query

```
SELECT s.name, c.name as Career, s.grade
FROM student AS s
JOIN career AS c
ON s.career_fk = c.id
ORDER BY s.grade DESC
OFFSET (7-1)*10
LIMIT 10;
```



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- Free space management is important for updates and inserts.
- Fragmentation management ensures efficient use of space.
- Data compression can also improve storage efficiency.



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.



Block and Page Organization

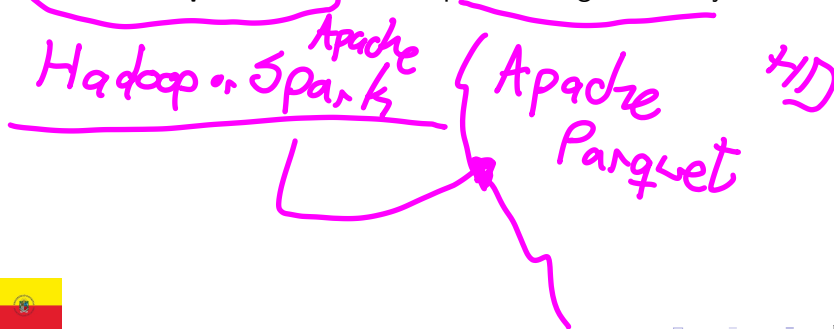
- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.

Distributed
Data System



Block and Page Organization

- Data is stored in **blocks** (pages) on disk.
- **Block size** and **layout** affect **I/O performance**.
- Records may be **packed**, **slotted**, or may **span multiple blocks**.
- **Free space management** is important for **updates** and **inserts**.
- **Fragmentation management** ensures efficient use of space.
- **Data compression** can also improve storage efficiency.



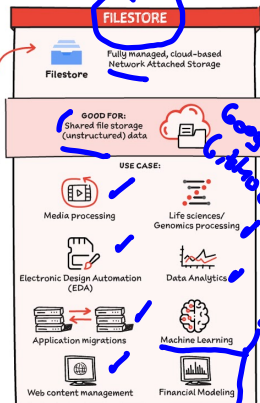
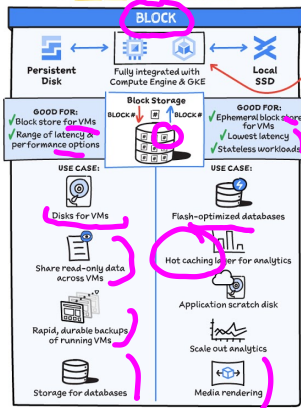
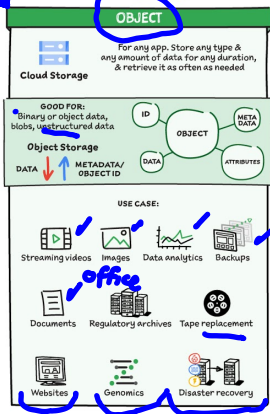
Record Storage: Use Cases

#GCPSketchnote

@PVERGADIA
THECLOUDGIRL.DEV
04.23.2021



Which Storage Should I Use?



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System**
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



Transactional System Concepts

- A **transaction** is a sequence of operations performed as a **single logical unit of work.** *queries*

- Transactions must satisfy the **ACID** properties:

- Atomicity: All or nothing.
- Consistency: Preserves database integrity.
- Isolation: Transactions do not interfere.
- Durability: Results persist after completion.

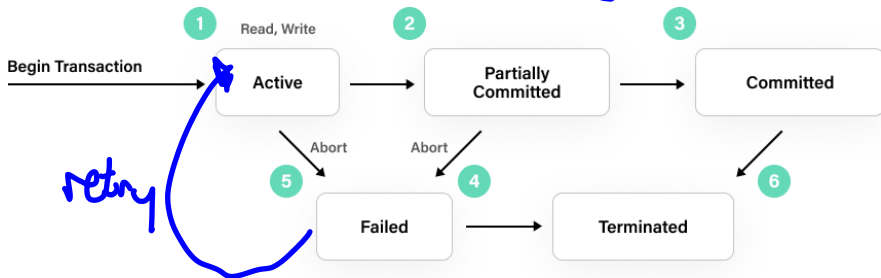
queue

Concurrency



Transaction Lifecycle

- **Begin**: Transaction starts. *→ enter*
- **Read/Write**: Operations are performed. *→ memory*
- **Commit**: Changes are made permanent. *← memory*
- **Rollback**: Changes are undone if an error occurs.
- **Savepoints** can be used for partial rollbacks. *↪*



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution**
- 6 Concurrency Control
- 7 Failure Recovery



Query Execution Process

- **Query execution** is the process of **interpreting** and **running** database queries.
- Steps:
 - **Parsing**: Analyzing query **syntax**.
 - **Optimization**: Choosing the best execution plan.
 - **Execution**: Retrieving and processing data.
- **Efficient execution** is critical for **performance**.

Snowflake

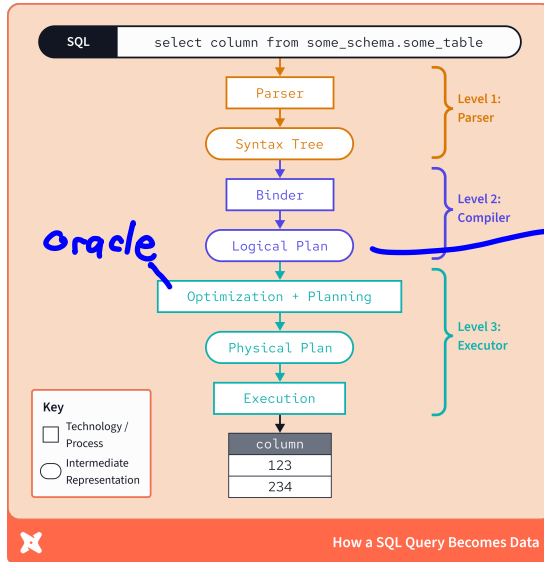
Compiler

SQL
functions

declarative



Query Execution Flow: Full Transaction



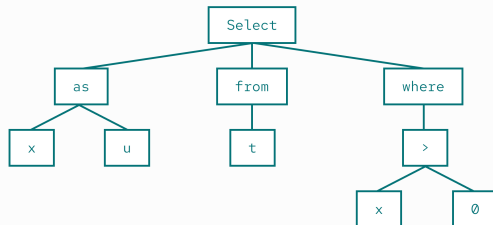
Query Execution Flow: Syntax Tree

SQL

`select x as u from t where x > 0`

Parser

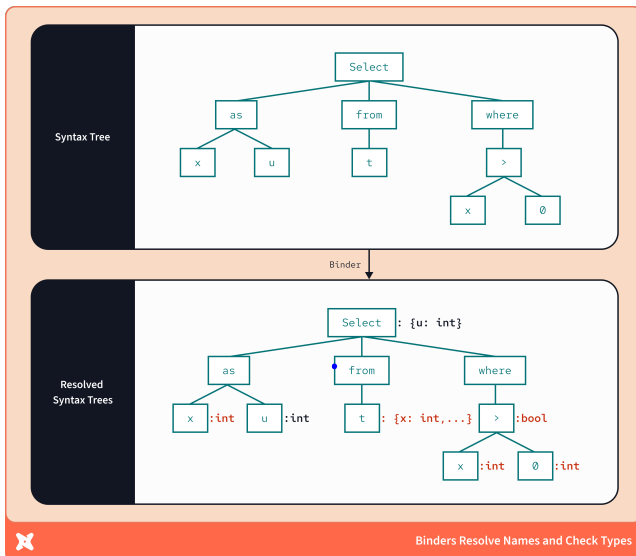
Syntax Tree



Parsers Recognize the Structure of the Query

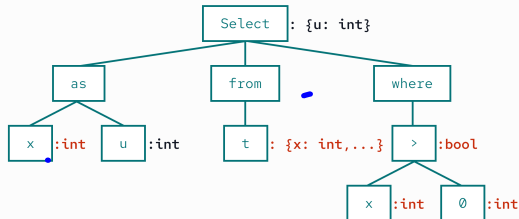


Query Execution Flow: Compilation



Query Execution Flow: Logical Plan

Resolved
Syntax Trees



Generating Logical Plan

Logical Plan

1. TableScan: t
2. Filter: t.x > 0
3. Projection: t.x as u



Compilation Produces an Executable Plan from a Resolved Syntax Tree



Query Optimization

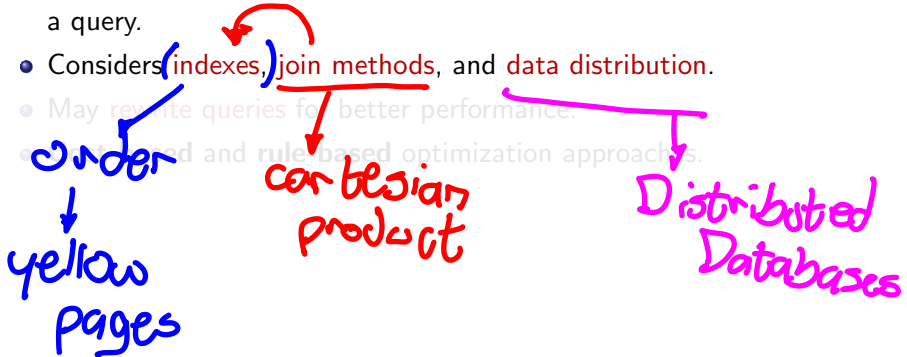
- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers indexes, join methods, and data distribution.
- May rewrite queries for better performance.
- Cost-based and rule-based optimization approaches.

resources
time memory



Query Optimization

- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers (indexes,) join methods, and data distribution.
- May rewrite queries for better performance.
- Cost-based and rule-based optimization approaches.



Query Optimization

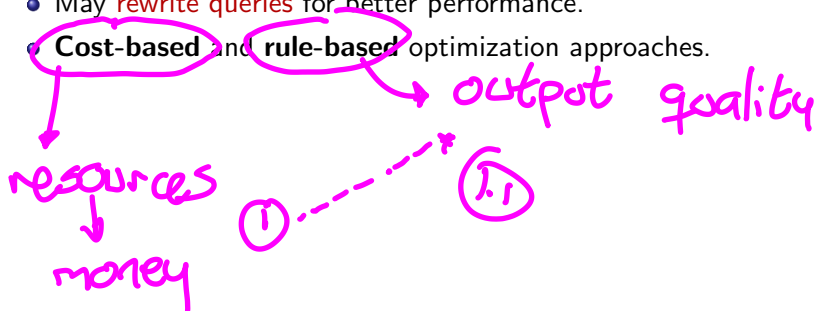
- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers **indexes**, **join methods**, and **data distribution**.
- May **rewrite queries** for better performance.
- Cost-based and rule-based optimization approaches.

internal → no the user

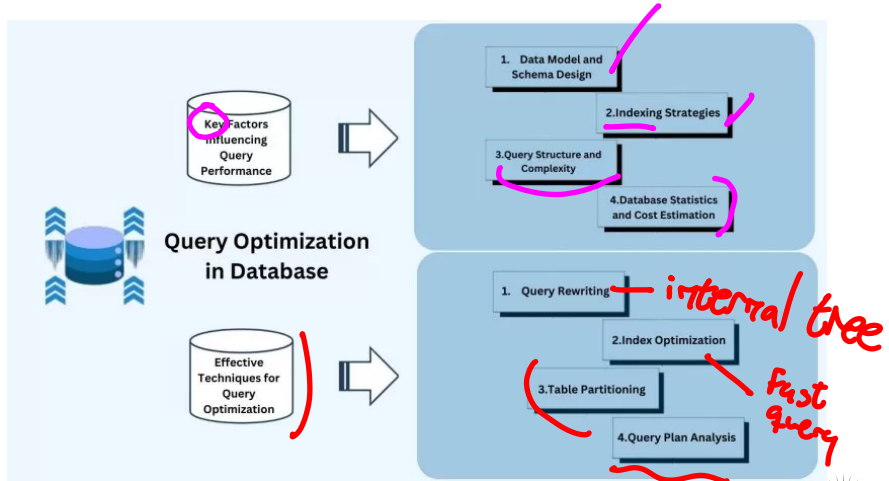


Query Optimization

- The **query optimizer** selects the most efficient strategy for executing a query.
- Considers **indexes**, **join methods**, and **data distribution**.
- May **rewrite queries** for better performance.
- **Cost-based** and **rule-based** optimization approaches.



Query Optimization Factors



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control**
- 7 Failure Recovery



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:

- Locking protocols: Use locks to control access (e.g., two-phase locking).
- Timestamp ordering: Assigns timestamps to transactions.
- Optimistic concurrency control: Assumes no conflicts.
- Multiversion concurrency control: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques.

• **Locking protocols:** Use locks to control access (e.g. two-phase locking).

- Timestamp ordering: Assigns timestamps to transactions.
- *Optimistic concurrency control:* Assumes no conflicts.
- *Pessimistic concurrency control:* Assumes conflicts may occur.

very

fast



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



Concurrency Control Concepts

- **Concurrency control** ensures correct results when **multiple transactions** execute **simultaneously**.
- Prevents problems like **lost updates**, **dirty reads**, and **deadlocks**.
- Techniques:
 - **Locking protocols**: Use locks to control access (e.g., two-phase locking).
 - **Timestamp ordering**: Assigns timestamps to transactions.
 - **Optimistic concurrency control**: Assumes no conflicts.
 - **Pessimistic concurrency control**: Assumes conflicts may occur.



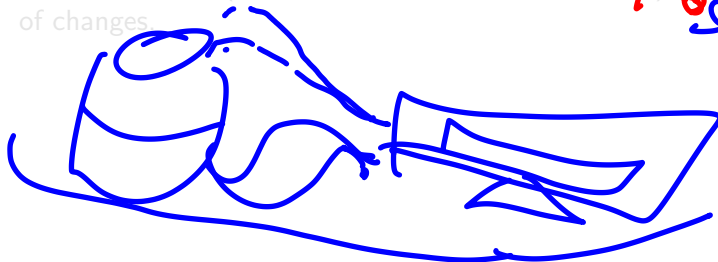
Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- Deadlocks occur when transactions wait indefinitely for each other.
- DBMS uses deadlock detection and resolution strategies.
- Isolation levels (e.g., Read Committed, Serializable) control visibility of changes.



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- DBMS uses **deadlock detection and resolution strategies**.
- **Isolation levels** (e.g., Read Committed, Serializable) control visibility of changes.



No progress!



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- **DBMS** uses **deadlock detection** and **resolution strategies**.
- **Isolation levels** (e.g., Read Committed, Serializable) control visibility of changes.



Locking and Deadlocks

- **Locks** prevent **conflicting operations** on the same data.
- **Deadlocks** occur when transactions **wait indefinitely** for each other.
- DBMS uses **deadlock detection** and **resolution strategies**.
- **Isolation levels** (e.g., Read Committed, Serializable) control visibility of changes.



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery**



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a **failure**.

- Types of failures:

- Transaction failure: Only one transaction fails.
- System crash: All active transactions are lost.
- Media failure: Disk or storage device fails.

- Recovery techniques:

- Write-ahead logging (WAL):
 - Log created before writing data.
 - Ensures durability and atomicity.
 - Periodically saves the database state.
 - Can be very fast.

- Checkpointing:
 - Periodically saves the database state.
 - Can be very slow.
 - Allows fast recovery to consistent state.



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.

- Types of failures:

- **Transaction failure**: Only one transaction fails.
- **System crash**: All active transactions are lost.
- **Media failure**: Disk or storage device fails.

- Recovery techniques:

• Write-ahead logging (WAL)

• Checkpointing

• Two-phase commit (2PC)

• Distributed transaction recovery

• Replication

• Backup and recovery

• Database recovery

• Database recovery



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.

Recovery techniques:

Write-ahead logging (WAL):

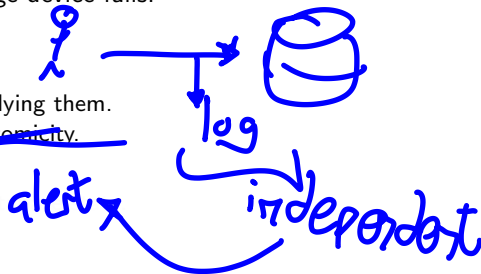
- Logs changes before applying them.
- Ensures durability and atomicity.

Checkpointing:

- Periodically saves the database to disk.
- Reduces recovery time.

Shadow paging:

- Maintains copies of data pages.
- Allows quick recovery to a consistent state.



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
- Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.
- Recovery techniques:
 - **Write-ahead logging (WAL)**:
 - Logs changes before applying them.
 - Ensures durability and atomicity.
 - **Checkpointing**:
 - Periodically saves the database state.
 - Reduces recovery time.
 - **Shadow paging**:
 - Maintains copies of data pages.
 - Allows quick recovery to a consistent state.

→ snapshot
↓
incremental
recovery



Failure Recovery Concepts

- **Failure recovery** restores the database to a **consistent state** after a failure.
 - Types of failures:
 - **Transaction failure**: Only one transaction fails.
 - **System crash**: All active transactions are lost.
 - **Media failure**: Disk or storage device fails.
 - Recovery techniques:
 - **Write-ahead logging (WAL)**:
 - Logs changes before applying them.
 - Ensures durability and atomicity.
 - **Checkpointing**:
 - Periodically saves the database state.
 - Reduces recovery time.
 - **Shadow paging**.
 - Maintains copies of data pages.
 - Allows quick recovery to a consistent state.
- Handwritten notes:* "Performance" with an arrow pointing to "Shadow paging", and "fast" with an arrow pointing to "Allows quick recovery to a consistent state".



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- **Recovery** ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- **Recovery** ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- Recovery ensures atomicity and durability.



Write-Ahead Logging and Checkpoints

- **Write-ahead logging (WAL):** All changes are **logged before** being applied to the database.
- **Checkpoints:** Periodically **save** the database **state** to speed up recovery.
- **Shadow paging:** Maintains **copies** of data **pages**.
- **Recovery ensures atomicity and durability.**



Outline

- 1 DataBases Infrastructure
- 2 DBMS Architecture
- 3 Database System Administration
- 4 Transactional System
- 5 Query Execution
- 6 Concurrency Control
- 7 Failure Recovery



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/databases-ii>

